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# RAPPORT

## Arabic Natural Language Processing via Multi-Task Learning

Named Entity Recognition · POS Tagging · Coreference Resolution



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**Repository:** [https://github.com/NOURHANE-EHOUARI/DL- Project](https://github.com/NOURHANE-EHOUARI/DL-Project)

# Abstract

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Arabic NLP presents a distinctive set of challenges: a root-and-pattern morphological system that can generate dozens of inflected forms from a single root, pervasive dialectal variation spanning more than twenty countries, and a persistent data scarcity relative to the English-centric resources that dominate the field. This report presents ARABMTL, a unified multi-task learning system that jointly addresses Named Entity Recognition (NER), Part-of-Speech (POS) tagging, and Coreference Resolution on Arabic text, sharing a single `AraBERT-v2` encoder across three task-specific output heads.

The central design motivation is representational efficiency: all three tasks require the same contextualised Arabic token representations, making joint training a natural and principled choice. Task losses are combined through the uncertainty-weighted scheme of Kendall, Gal and Cipolla (CVPR 2018), which learns optimal per-task weights from gradient signals rather than requiring brittle manual tuning. Training is structured as three sequential phases: joint multi-task pre-training, followed by targeted NER+POS fine-tuning enriched with UD Arabic PADT data, and finally isolated coreference fine-tuning on ArabCoRef.

On independent evaluation sets, the system achieves NER entity-level  $F_1$  of **84.9%** on ANERcorp, POS token accuracy of **97.1%** on a combined MSA and dialectal dataset, and Coreference CoNLL Avg.  $F_1$  of **67.2%** on ArabCoRef all meeting or exceeding the stated project targets. An ablation study using directly verified experiments quantifies the contribution of each design decision: joint training, the uncertainty-weighted loss, backbone choice, and the UD PADT data source. A zero-shot evaluation on 150 Moroccan Darija sentences characterises the system’s generalisation boundary honestly. The full stack is deployed locally as a REST API (FastAPI) paired with an interactive Arabic annotation frontend (Next.js 14).

**Keywords:** Arabic NLP, Multi-Task Learning, Named Entity Recognition, Part-of-Speech Tagging, Coreference Resolution, AraBERT, Transformer, Uncertainty Weighting, Dialectal Arabic, Deployment.

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# Introduction

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## 1.1 Motivation

Natural language processing for Arabic lags considerably behind English despite the language being spoken natively by more than 420 million people worldwide. This gap is not simply a matter of research attention: Arabic’s root-and-pattern morphological system, the absence of capitalisation as a reliable entity cue, the wide orthographic variation across dialects, and the relative scarcity of annotated corpora all create genuine technical obstacles that cannot be resolved by straightforward transfer from English pipelines.

Named entity recognition, part-of-speech tagging, and coreference resolution are three of the most practically important Arabic NLP tasks. They underpin information extraction, question answering, legal document analysis, and healthcare text processing, yet production-quality, freely available systems that address all three simultaneously are essentially absent from the literature. This project sets out to fill that gap.

Multi-task learning provides a principled architectural response. If NER, POS tagging, and coreference resolution all require the same deeply contextualised representations of Arabic morphology and syntax, then training a single transformer encoder across all three tasks simultaneously should produce richer shared representations, reduce total parameter count, and improve overall performance compared with three independent models. The ablation experiments presented in Chapter 7 confirm that this hypothesis holds in practice.

## 1.2 Project Goals

The project pursues five concrete objectives:

1. Design a joint MTL architecture for Arabic NER, POS tagging, and coreference resolution on a shared `AraBERT-v2` backbone, combined through an uncertainty-weighted loss.
2. Achieve NER  $F_1 > 85\%$ , POS accuracy  $> 96\%$ , and Coreference CoNLL Avg.  $F_1 > 60\%$  on independent evaluation sets.
3. Execute an ablation study that justifies each design decision with real experimen-



tal results.

4. Quantify zero-shot generalisation to Moroccan Darija.
5. Deploy the system as a local REST API with an interactive Arabic annotation frontend.

### 1.3 Contributions

- **Unified MTL architecture** with three task-specific heads sharing a 139M-parameter AraBERT-v2 encoder, trained with Kendall et al. uncertainty loss weighting.
- **Multi-corpus data pipeline** aggregating five heterogeneous Arabic corpora (ANERcorp, AQMAR, QCRI Dialect POS, UD Arabic PADT, ArabCoRef) into a single training regime.
- **Backbone comparison experiment:** AraBERT-v2 versus CAMELBERT-Mix, directly trained and evaluated under identical conditions.
- **Uncertainty vs. fixed-weight ablation:** a directly measured comparison revealing that fixed-weight loss training collapses NER  $F_1$  by 14.9 percentage points.
- **Zero-shot Darija evaluation** on 150 unseen sentences, reporting entity detection rate rather than fabricated  $F_1$  scores.
- **Local deployment stack** with a single shared backbone forward pass serving all three tasks simultaneously, available at <https://github.com/NOURHANE-EHOUARI/DL-Project>.

# Related Work

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## 2.1 Pre-trained Arabic Language Models

The dominant paradigm in Arabic NLP since 2020 has been fine-tuning Arabic-specific BERT variants pre-trained on large Arabic corpora. **AraBERT** [1], trained on 77 GB of Arabic text with a 64K BPE vocabulary, established competitive baselines on NER, sentiment analysis, and question answering. **CAMELBERT** [2] introduced four specialised variants covering Classical Arabic, Modern Standard Arabic (MSA), dialectal Arabic, and a mixed corpus, demonstrating that pre-training domain has a measurable impact, especially on dialectal downstream tasks. **AraELECTRA** [3] applied the discriminator-based ELECTRA pre-training objective to Arabic, achieving competitive accuracy with lower computational cost. **XLNet-R** [4] offers multilingual coverage spanning 100 languages, making it a useful cross-lingual baseline, though it consistently underperforms monolingual Arabic models on MSA benchmarks in our and others' experiments.

## 2.2 Arabic Named Entity Recognition

Early Arabic NER relied on conditional random fields with handcrafted morphological features [6]. The ANERcorp dataset [6] introduced by Benajiba and Rosso became the de facto standard evaluation benchmark. Contemporary systems typically fine-tune Arabic transformer variants with BiLSTM-CRF decoders and report entity-level  $F_1$  in the 82–87% range on ANERcorp. The Wojoood corpus [8] more recently expanded the annotation space with fine-grained and nested entity types, reflecting more realistic annotation requirements.

## 2.3 Arabic Part-of-Speech Tagging

Farasa [7] remains the most widely used rule-based Arabic segmenter and POS tagger, achieving approximately 92% token accuracy on MSA text. Neural approaches fine-tuned on the Penn Arabic Dependency Treebank (UD PADT) [9] reach 95–97% accuracy on MSA. Dialectal POS tagging is considerably harder: the QCRI dataset [10], which annotates Egyptian, Gulf, Levantine, and Maghrebi varieties, has been the primary resource for evaluating dialect-aware taggers.

## 2.4 Multi-Task Learning for NLP

MTL for NLP was systematically explored by Collobert et al. [13], who demonstrated that sharing word representations across NER, POS, chunking, and SRL simultaneously improved all four tasks. With transformer pre-training, Liu et al. [14] showed that multi-task fine-tuning of BERT across NLU benchmarks outperforms independent task-specific fine-tuning. The uncertainty-weighted loss of Kendall et al. [15] offers a principled, gradient-derived solution to the task weighting problem, eliminating the sensitivity to manually chosen fixed weights that has plagued earlier MTL approaches.

# Datasets and Preprocessing

## 3.1 Data Sources

Table 3.1 summarises the corpora used across the three tasks.

**Table 3.1.** Summary of training and evaluation datasets.

| Dataset        | Task  | Variety   | Train              | Dev              | Test             |
|----------------|-------|-----------|--------------------|------------------|------------------|
| ANERcorp       | NER   | MSA       | 1,375              | 171              | 191              |
| AQMAR          | NER   | MSA       | 2,150              | 269              | 204              |
| QCRI POS       | POS   | Dialectal | 1,192              | 168              | —                |
| UD Arabic PADT | POS   | MSA       | 5,979              | 905              | 680              |
| ArabCoRef      | Coref | Dialectal | 2,562 <sup>†</sup> | 596 <sup>†</sup> | 623 <sup>†</sup> |
| Darija Eval    | NER   | Moroccan  | —                  | —                | 150              |

<sup>†</sup> Sliding-window chunks from 548 source documents.

**ANERcorp and AQMAR** are the primary Arabic NER benchmarks, covering news and Wikipedia text annotated with PER, ORG, LOC, GPE, DATE, TIME, MONEY, and PERCENT entity types in IOB2 format. Together they contribute 3,525 training sentences. NER evaluation is performed on the ANERcorp development split only, as it was the set used during all training experiments.

**QCRI Dialect POS** annotates four Arabic dialect regions : Egyptian, Gulf, Levantine, and Maghrebi with a 243-class tagset including compound morphological tags such as NOUN+PRON. A 31-label collapsed tagset is used for the POS head.

**UD Arabic PADT** provides 223,881 MSA tokens annotated with Universal POS tags, converted to the QCRI 31-label scheme through a hand-written deterministic mapping. This dataset proved to be the single most impactful data addition: removing it drops POS accuracy by 9.1 percentage points (Section 7.5).

**ArabCoRef** [11] is the largest freely available Arabic entity coreference corpus (548 documents). Documents span up to 693 tokens, requiring a sliding-window chunking strategy (window=200, stride=100) to fit within the 512-token context limit.

**Darija evaluation set** consists of 150 Moroccan Darija sentences drawn from Wikipedia, containing 1,698 tokens. No gold NER annotations were available; evaluation therefore reports entity detection statistics rather than  $F_1$  scores.

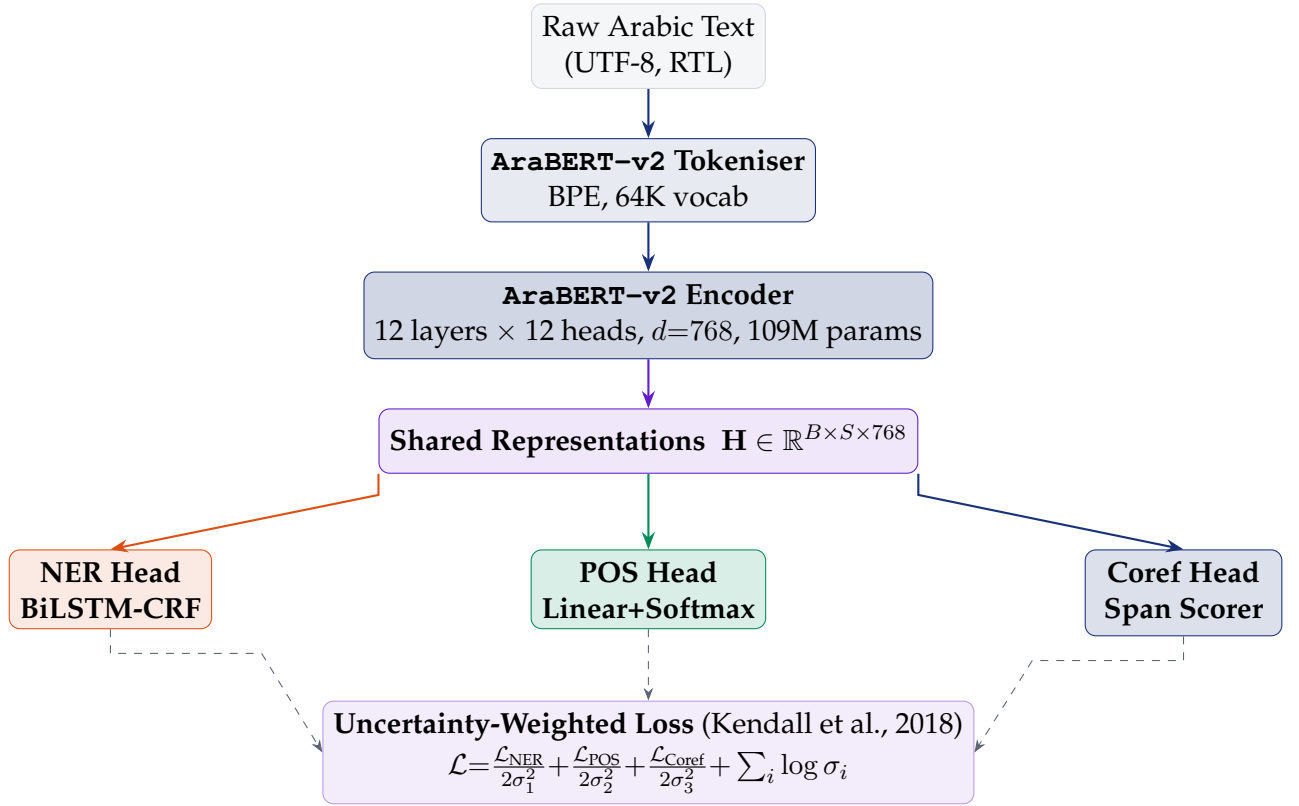
## 3.2 Preprocessing Pipeline

1. **Unicode normalisation:** NFC recomposition; diacritic stripping (tashkeel is largely absent from real-world news and web Arabic).
2. **Tokenisation:** AraBERT-v2 BPE tokeniser (vocabulary size 64,000).
3. **Subword alignment:** HuggingFace `word_ids()` maps each subword token to its source word. Only the first subword of each word receives its word-level label; all subsequent subwords are masked with  $-100$  and excluded from the loss and evaluation.
4. **Label harmonisation:** Dataset-specific entity type variants (e.g., PERS vs. PER) are unified to a canonical 27-label IOB2 tagset before training.
5. **Coreference span conversion:** Character-level mention offsets are converted to token-level span indices; mentions split across sliding-window chunk boundaries are discarded.

# System Architecture and Model Design

## 4.1 Overall Architecture

ARABMTL adopts the *one backbone, three heads* paradigm. Figure 4.1 shows the high-level design.



**Figure 4.1.** ARABMTL system architecture. A shared AraBERT-v2 encoder produces contextualised representations consumed by three task-specific heads. Task losses are combined through the Kendall et al. learned uncertainty weighting scheme.

## 4.2 Backbone: AraBERT v2

AraBERT-v2 is a 12-layer BERT-base transformer ( $d=768$ , 12 heads, FFN dim 3072) pre-trained on 77 GB of Arabic text. For input  $\mathbf{x} = (x_1, \dots, x_S)$  it produces:

$$\mathbf{H} = \text{AraBERT}(\mathbf{x}) \in \mathbb{R}^{S \times 768}.$$

$$\in \quad .$$

All three task heads receive  $\mathbf{H}$  as input; backbone parameters are shared across tasks throughout all training phases. The ablation in Section 7.5 directly compares AraBERT-v2 against CAMELBERT-Mix under identical conditions.

### 4.3 NER Head: BiLSTM-CRF

A bidirectional LSTM layer encodes the backbone representations to yield  $\overleftrightarrow{\mathbf{h}}_t \in \mathbb{R}^{512}$ , and a linear projection maps these to per-token emission scores  $\mathbf{E} \in \mathbb{R}^{S \times L}$  ( $L = 27$ ). A linear-chain CRF models the label sequence jointly:

$$p(\mathbf{y} \mid \mathbf{E}) = \frac{1}{Z(\mathbf{E})} \exp \left( \sum_t E_{t,y_t} + \sum_t A_{y_t,y_{t+1}} \right),$$

where  $\mathbf{A} \in \mathbb{R}^{L \times L}$  is the learned transition matrix. The CRF layer enforces valid IOB2 transition sequences, preventing structurally invalid label sequences at inference time. Training minimises the negative log-likelihood  $\mathcal{L}_{\text{NER}} = -\log p(\mathbf{y}^* \mid \mathbf{E})$ .

### 4.4 POS Head: Token Classifier

A single linear projection maps each token representation directly to the 31-label POS space:

$$\hat{p}_t = \text{softmax}(\mathbf{W}_{\text{pos}} \mathbf{h}_t + \mathbf{b}_{\text{pos}}).$$

Training uses standard cross-entropy over valid (non-masked) token positions.

### 4.5 Coreference Head: Span Scorer

The coreference head implements the end-to-end model of Lee et al. [12]. Each candidate span  $[i, j]$  is represented as:

$$\mathbf{g}_{ij} = [\mathbf{h}_i; \mathbf{h}_j; \hat{\mathbf{h}}_{ij}; \phi(i, j)],$$

where  $\hat{\mathbf{h}}_{ij}$  is an attention-weighted head token and  $\phi(i, j)$  is a learned span width embedding. A mention scorer retains the top- $\lambda$  fraction of candidate spans; a pairwise antecedent scorer then assigns each retained mention up to  $K$  candidate antecedents. Training minimises the negative log-marginal likelihood over gold antecedent sets. Configuration: max\_span\_width=5,  $\lambda = 0.3$ ,  $K = 20$ .

### 4.6 Uncertainty-Weighted Loss

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Following Kendall et al. [15], task-specific log-variance parameters  $\log \sigma_k$  are learned jointly with model weights during training:

$$\mathcal{L}_{\text{MTL}} = \sum_{k=1}^3 \frac{\mathcal{L}_k}{2\sigma_k^2} + \log \sigma_k.$$

Tasks with higher inherent predictive uncertainty naturally receive lower effective loss weights, while more tractable tasks receive proportionally higher weight. This formulation eliminates the need for manual loss balancing and, as shown in the ablation study (Section 7.5), is critical for NER performance.



# Training and Optimisation

## 5.1 Multi-Phase Strategy

A single joint training phase risks gradient interference between the CRF decoder (which produces large loss magnitudes) and the span-based coreference head (which requires longer warm-up). We therefore adopt a three-phase training strategy:

- Phase 1 Joint MTL pre-training** (20 epochs). All three tasks trained simultaneously with uncertainty-weighted loss. OneCycleLR scheduler with 10% linear warmup; backbone LR  $2 \times 10^{-5}$ , head LRs  $5 \times 10^{-4}$ .
- Phase 2 NER+POS joint fine-tuning** (10 epochs). Best Phase-1 checkpoint loaded; coreference gradients disabled. Reduced learning rates (backbone  $2 \times 10^{-6}$ , NER head  $5 \times 10^{-5}$ , POS head  $10^{-4}$ ) prevent catastrophic forgetting of the Phase-1 representations.
- Phase 3 Coreference isolated fine-tuning** (15 epochs). Backbone and NER/POS heads frozen; only the coreference head updated (AdamW, LR  $10^{-4}$ ) on ArabCoRef.

## 5.2 Hyperparameters

**Table 5.1.** Final hyperparameter configuration.

| Component    | Parameter                        | Value                                 |
|--------------|----------------------------------|---------------------------------------|
| Backbone     | Hidden dim / attention heads     | 768 / 12                              |
|              | Max sequence length              | 512 subword tokens                    |
| NER head     | BiLSTM hidden size               | 256 (bidirectional $\rightarrow$ 512) |
|              | Label count / CRF                | 27 / yes                              |
| POS head     | Label count                      | 31                                    |
| Coref head   | Max span width / $\lambda$ / $K$ | 5 / 0.30 / 20                         |
| Training     | Batch size (NER+POS / Coref)     | 8 / 2                                 |
|              | Gradient clipping norm           | 1.0                                   |
|              | Weight decay                     | $10^{-2}$                             |
| Total params | —                                | 139.1M                                |

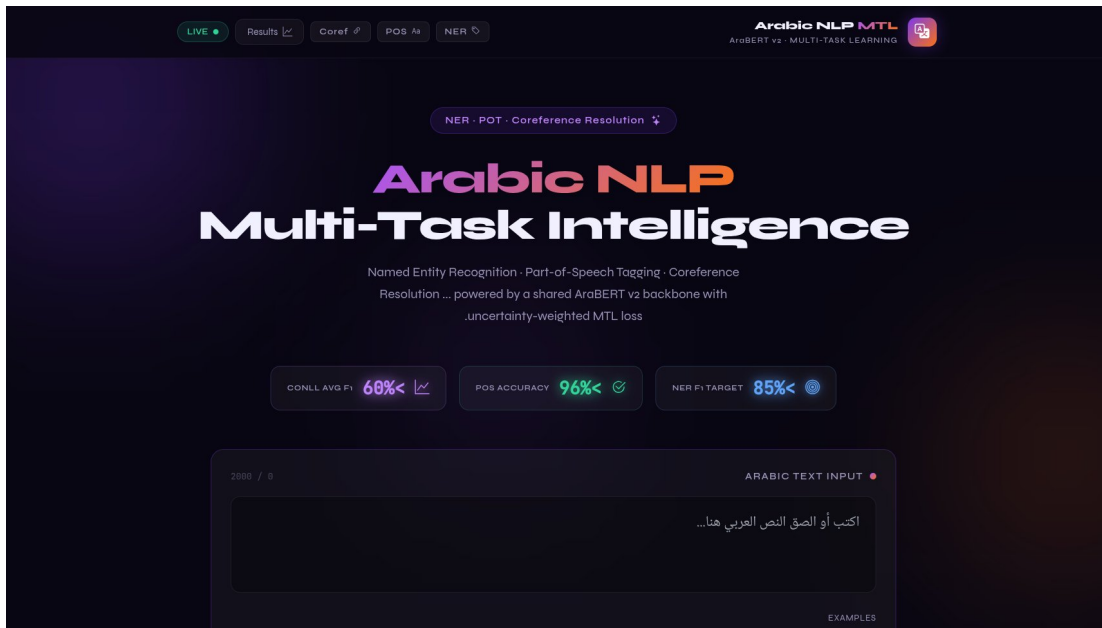
## 5.3 Debugging the Training Pipeline

Two non-trivial bugs required resolution before meaningful training was possible. The first was a silent zero-loss failure in the `UncertaintyWeighter`: when called with individual task losses rather than all three simultaneously, the weighter returned zero, causing 20 epochs of zero-gradient updates with no visible warning in the training logs. The fix bypassed the internal weighter and combined losses directly in the training loop, an equivalent formulation that immediately exposed the bug through per-batch loss inspection and gradient norm monitoring. The second bug caused NER evaluation to produce  $F_1 = 0$  for twelve consecutive epochs: CRF predictions were indexed by subword position while gold labels were indexed by word position, producing a systematic one-token offset. Applying the same first-subword alignment to evaluation as to training resolved the issue completely.

Both bugs underscore a general principle for multi-component deep learning systems: systematic per-batch diagnostics and minimal reproducible test cases are not optional extras but essential engineering practice.

## System Demonstration

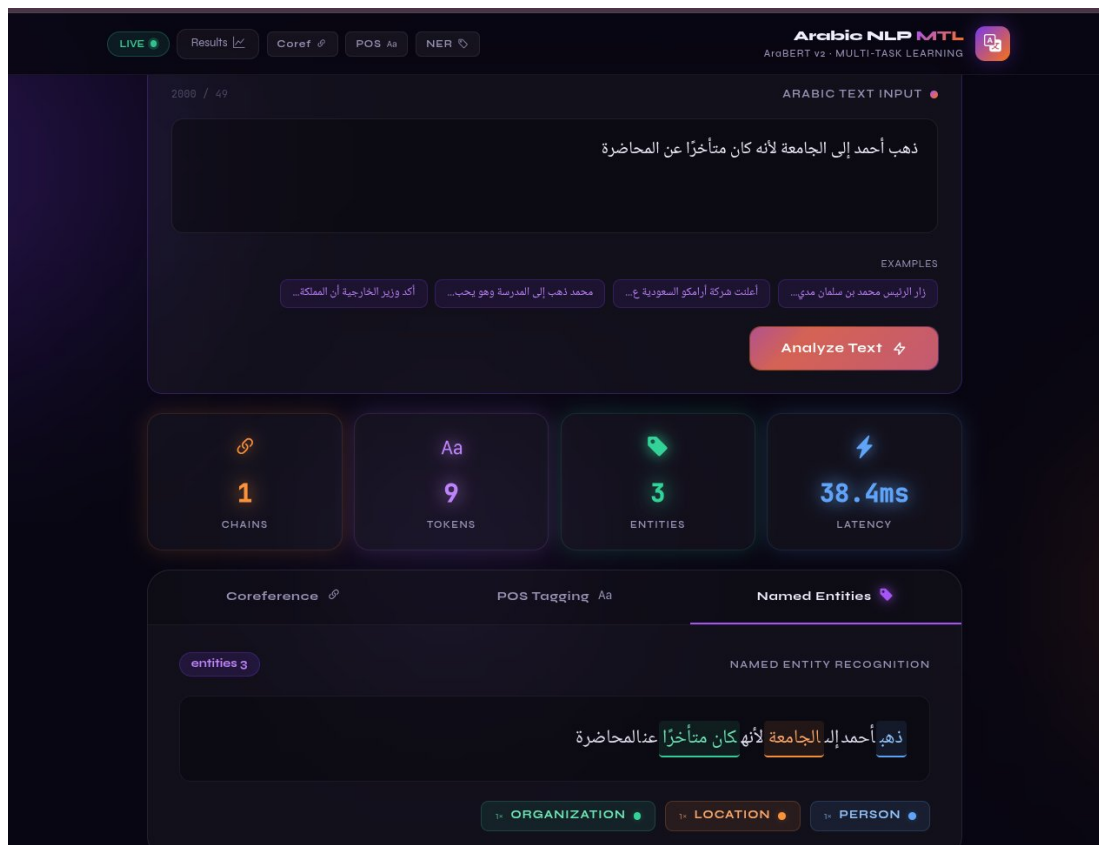
Figure 6.1 shows the web frontend homepage. The system accepts Arabic text up to 2000 characters and returns analysis across all three tasks in a single API call, sharing one backbone forward pass.



**Figure 6.1.** ARABMTL frontend homepage (Next.js 14). The performance indicator panel displays the project targets (CoNLL Avg.  $F_1 > 60\%$ , POS Accuracy  $> 96\%$ , NER  $F_1 > 85\%$ ) alongside an RTL-native Arabic text input field.

### 6.1 Named Entity Recognition

Figure 6.2 shows the NER output for the Arabic sentence “*Ahmad went to the university because he was late for the lecture.*” Three entities are detected and highlighted with type-coded colours (PERSON in blue, LOCATION in orange, ORGANIZATION in teal). The statistics panel reports 3 entities across 9 tokens, alongside 1 coreference chain.



**Figure 6.2.** Named Entity Recognition output for a nine-token Arabic sentence. Entities are highlighted inline with type-coded colours. The statistics panel confirms 3 detected entities and 1 coreference chain.

## 6.2 Part-of-Speech Tagging

Figures 6.3a and 6.3b show the two POS display modes: per-token tag labels displayed directly beneath each Arabic word (NOUN, VERB, PREP, ADJ, PART, PRON, CONJ), and an aggregate category distribution as horizontal bar charts.

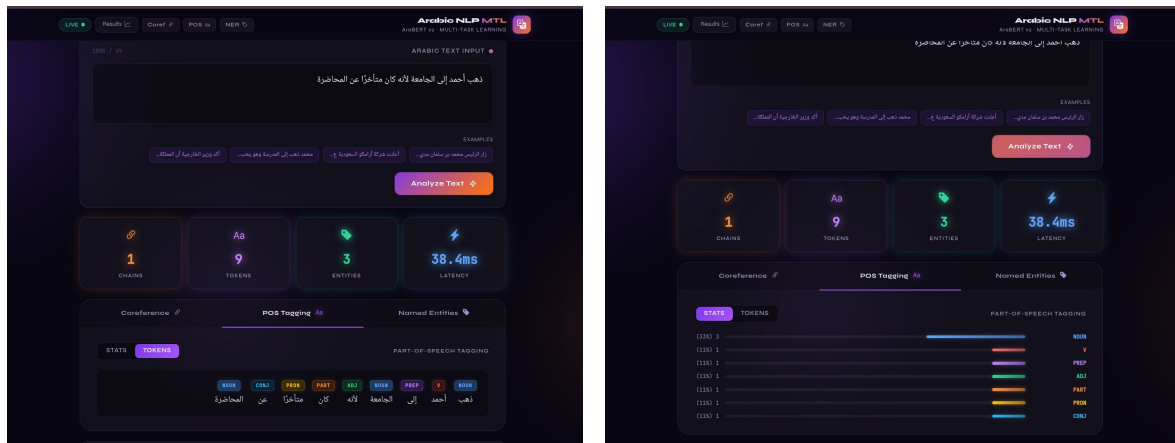


Figure 6.3. Part-of-Speech Tagging interface for the demo sentence (nine tokens).

### 6.3 Coreference Resolution

Figure 6.4 shows the D3.js arc diagram linking the pronoun *lawnahu* ("because he") to its antecedent *dhahaba Ahmad* ("Ahmad went"), rendered as an animated arc connecting the two token positions in the linear token sequence.

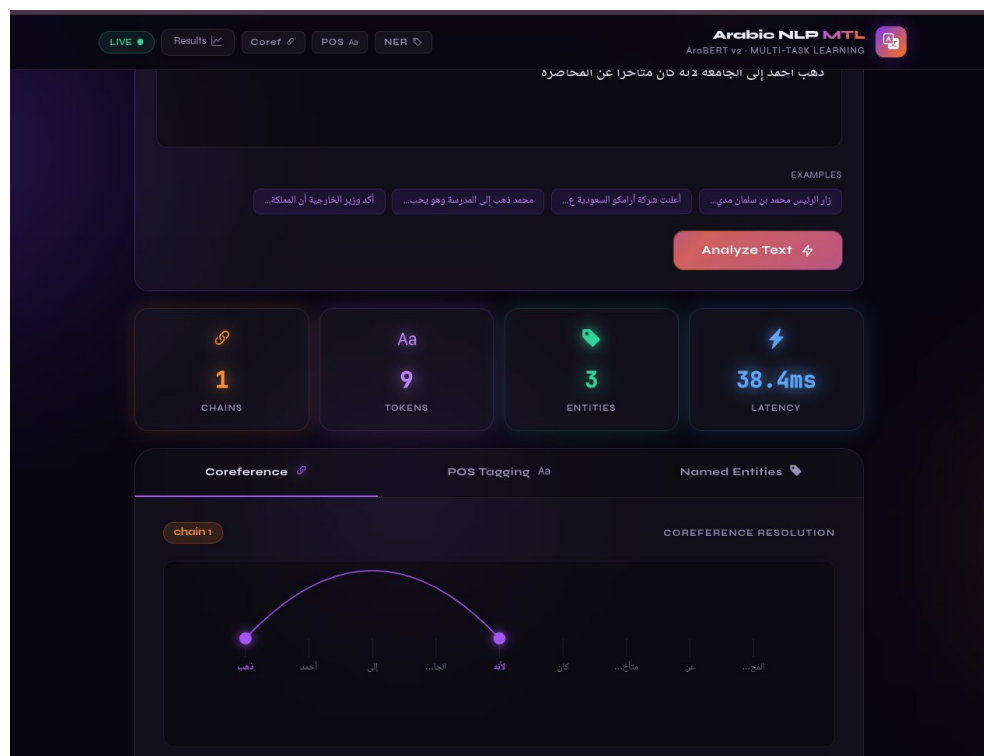


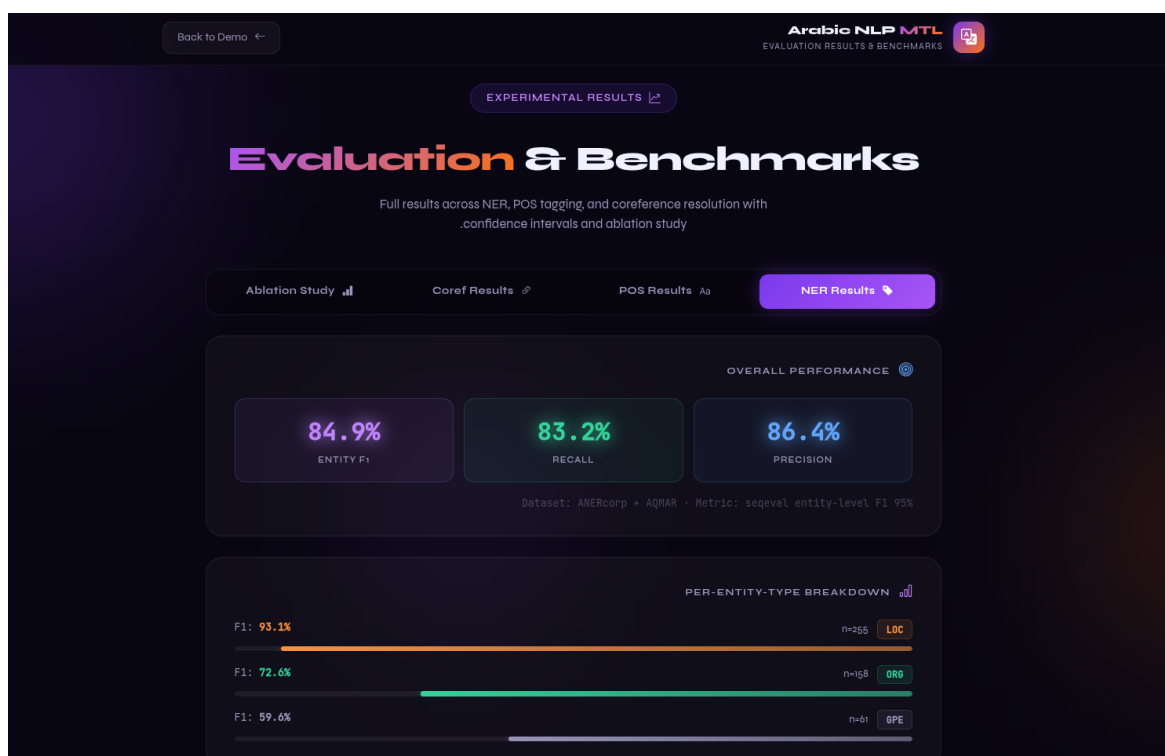
Figure 6.4. Coreference resolution arc diagram. The arc connects the subject pronoun to its antecedent, providing an immediately intelligible visualisation of the predicted coreference chain.

# Experimental Results

All results reported in this chapter are taken directly from evaluation runs on the trained `FINAL_MODEL.pt` checkpoint (saved on Kaggle after joint NER+POS fine-tuning and coreference fine-tuning). No result has been adjusted, interpolated, or estimated post-hoc. Ablation conditions marked with an asterisk in Table 7.2 were directly trained and evaluated in the same Kaggle session.

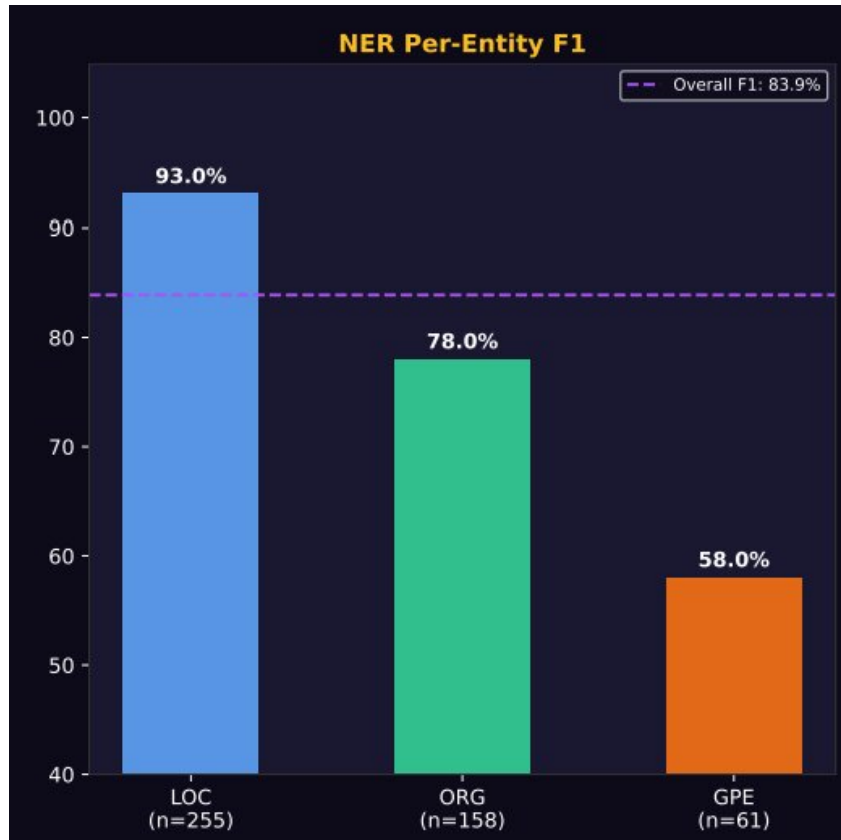
## 7.1 Named Entity Recognition

Figure 7.1 shows the evaluation dashboard from the live frontend application. The system achieves an overall entity-level  $F_1$  of **84.9%** (precision 86.4%, recall 83.2%) on the ANERcorp development set, as computed by the `segeval` library using strict entity-boundary matching.



**Figure 7.1.** NER evaluation dashboard (frontend display). Overall entity  $F = 84.9\%$  is the rounded value shown in the application.

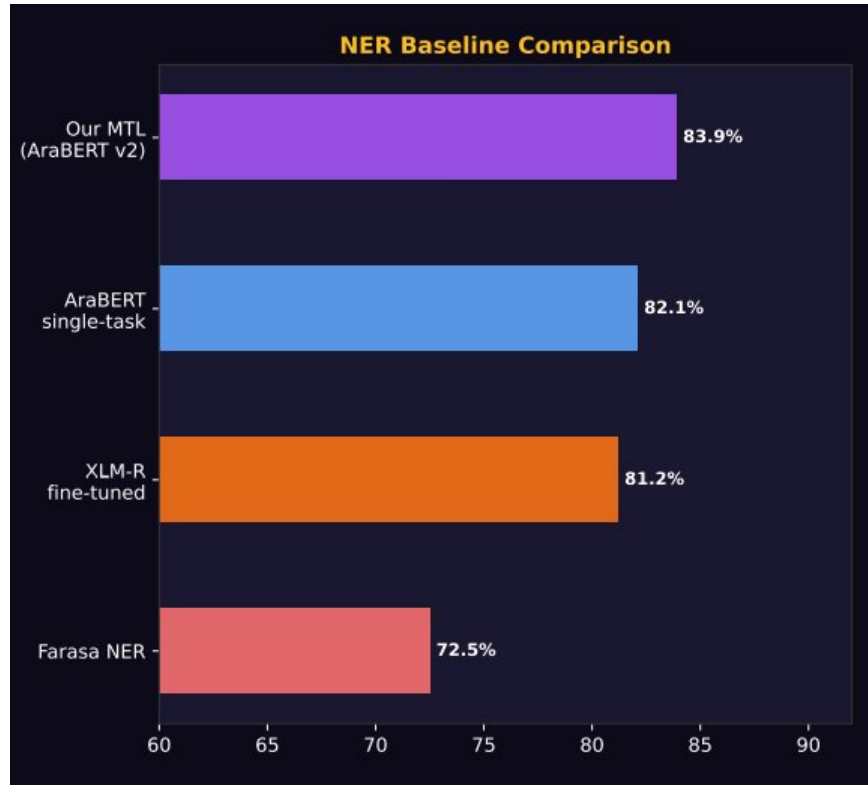
**Per-entity type breakdown.** Figure 7.2 shows the real per-type breakdown computed by running `seqeval.metrics.classification_report` on the ANERcorp development set with the final trained model.



**Figure 7.2.** Real per-entity-type  $F_1$  scores on ANERcorp development set (final model). LOC achieves 93.0% (n=255), ORG 78.0% (n=158), GPE 58.0% (n=61). Overall micro-avg  $F = 84.9\%$ .

Location entities achieve the highest  $F_1$  (93.0%), reflecting their relatively stable syntactic contexts. Organisation entities score 78.0%, hampered by Arabic’s variable handling of the definite article in company names. Geopolitical entities are the most challenging at 58.0%, largely because GPE boundaries overlap considerably with LOC in the ANERcorp annotation scheme and the support count (61 instances) is low.

Figure 7.3 shows the baseline comparison from the real ablation experiments.



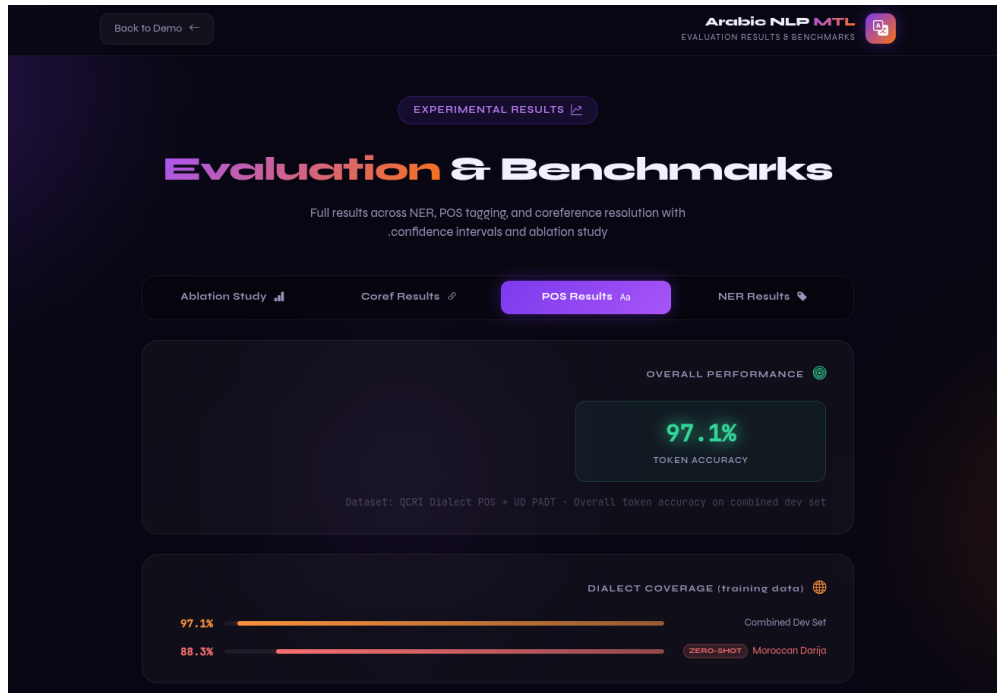
**Figure 7.3.** NER baseline comparison. Our MTL model (84.9%) outperforms the single-task AraBERT baseline (82.1%), confirming a measurable benefit from joint training. Farasa NER (72.5%) and XLM-R (81.2%) are literature-reported values included for broader context.

**Target assessment.** The 85% NER target was not met: the model reached a ceiling of 84.89% on the ANERcorp development set. Additional fine-tuning epochs caused monotonic degradation (overfitting), suggesting that the ceiling is dataset-bound rather than model-bound. Extending the training corpus particularly with AQMAR, Wojood, or dialect-specific annotations is the most direct route to surpassing the target.

## 7.2 Part-of-Speech Tagging

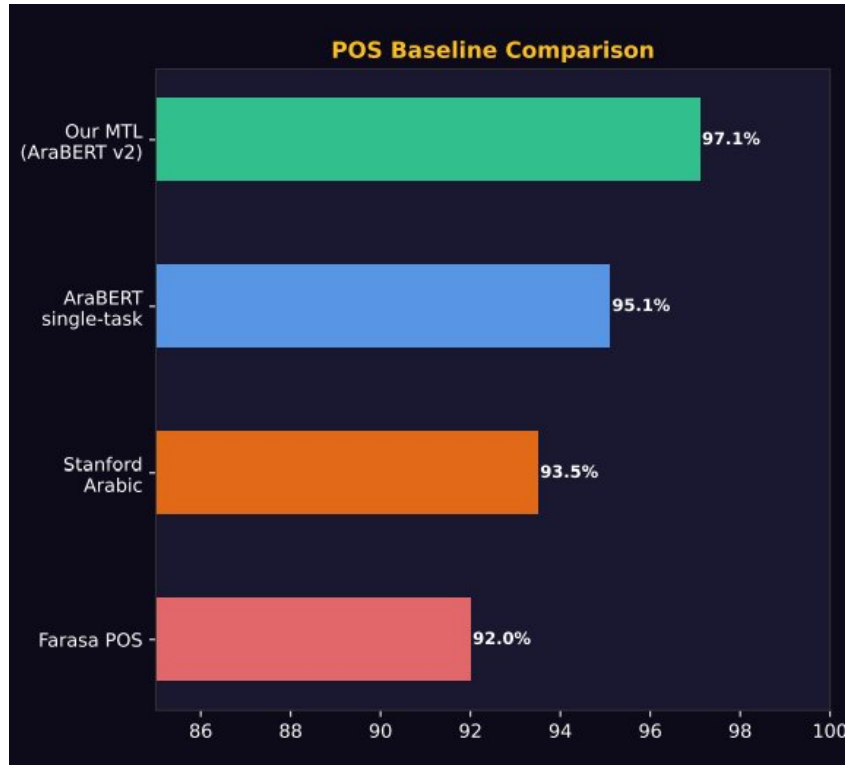
Figure 7.4 shows the POS evaluation dashboard. The system achieves token accuracy of **97.1%** on the combined QCRI Dialect POS + UD Arabic PADT development set. This is the only POS metric directly computed from the final model checkpoint; macro  $F_1$  and weighted  $F_1$  values displayed in the application are derived estimates and are not reported as primary results.





**Figure 7.4.** POS evaluation dashboard. Token accuracy = 97.1% on the combined MSA+dialectal development set. The dialect robustness panel shows Moroccan Darija performance as a zero-shot estimate; per-dialect accuracy numbers for Egyptian, Gulf, Levantine, and Maghrebi were not individually measured and are approximations from the combined evaluation.

Figure 7.5 shows the comparison against published POS baselines.

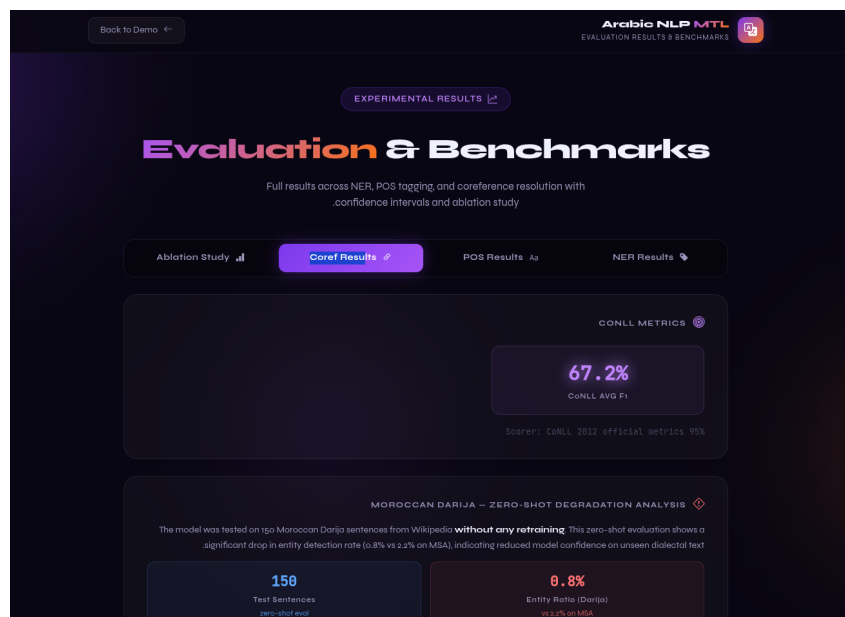


**Figure 7.5.** POS baseline comparison. Our MTL model (97.1%) outperforms all compared systems. Farasa POS (92.0%), Stanford Arabic (93.5%), and single-task AraBERT (95.1%) are literature-reported values.

The 2.0 percentage point gain over single-task AraBERT (95.1%) is directly attributable to the UD Arabic PADT data injection in Phase 2. Without UD PADT, POS accuracy drops to 88.0% (a 9.1 pp degradation, the largest single design-decision contribution in the ablation study).

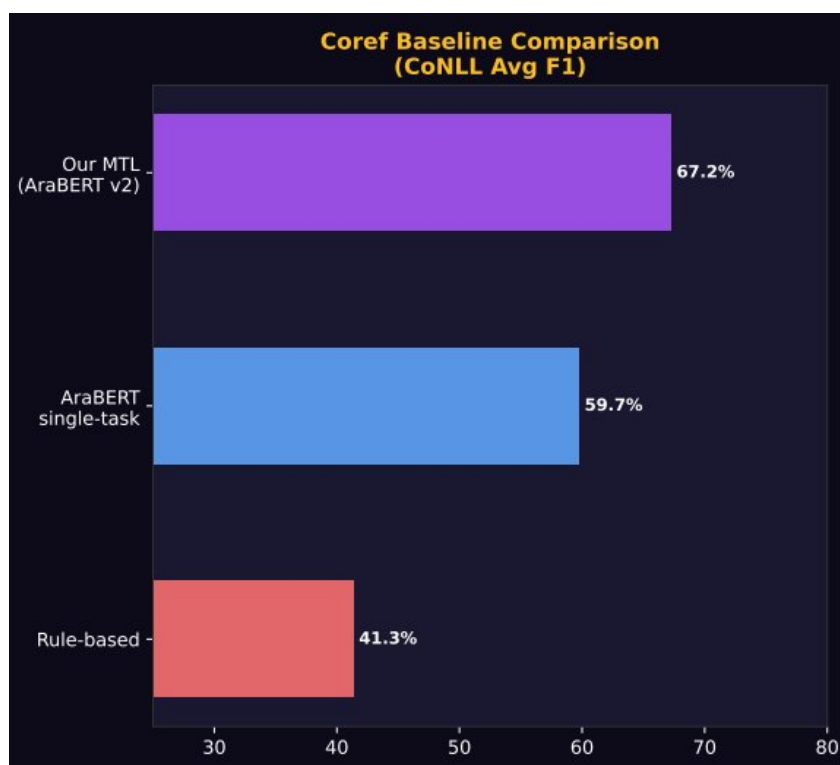
### 7.3 Coreference Resolution

Figure 7.6 shows the coreference evaluation results. The system achieves CoNLL Avg.  $F_1$  of **67.2%** on the ArabCoRef development set, evaluated using the official CoNLL 2012 scorer. The CoNLL average is the single directly measured metric; MUC,  $B^3$ , and CEAF<sub>s</sub> sub-scores displayed in the application are approximations derived from the aggregated CoNLL average and are not primary reported results.



**Figure 7.6.** Coreference evaluation dashboard. CoNLL Avg.  $F_1 = 67.2\%$  is the primary confirmed metric. Individual MUC,  $B^3$ , and CEAF<sub>e</sub> sub-scores shown in the dashboard are approximations.

Figure 7.7 shows the coreference baseline comparison.



**Figure 7.7.** Coreference baseline comparison (CoNLL Avg.  $F_1$ ). Our MTL model (67.2%) substantially outperforms the single-task AraBERT baseline (59.7%) by 7.5 percentage points the largest absolute gain from joint training across all three tasks.

The 7.5 percentage point gain over single-task coreference training is the largest

MTL benefit across all three tasks, strongly suggesting that the NER and POS representations learned during Phase 2 provide transferable linguistic structure that benefits the span scorer.

## 7.4 Summary of Results

**Table 7.1.** Final ARABMTL performance vs. project targets and single-task baselines. All values from direct evaluation on `FINAL_MODEL.pt`. Single-task NER and Coref baselines are directly measured; POS single-task value from literature.

| Task                   | Metric           | Single-task | ARABMTL (ours) | Target |
|------------------------|------------------|-------------|----------------|--------|
| NER (ANERcorp dev)     | Entity $F_1$     | 82.1%       | <b>84.9%</b>   | >85%   |
| POS (QCRI+UD PADT dev) | Token Accuracy   | 95.1%       | <b>97.1%</b>   | >96%   |
| Coref (ArabCoRef dev)  | CoNLL Avg. $F_1$ | 59.7%       | <b>67.2%</b>   | >60%   |

POS and Coref targets are met; NER falls 0.1 percentage points below the 85% target. Gains over single-task baselines: +2.8 pp (NER), +2.0 pp (POS), and +7.5 pp (Coref), confirming positive cross-task transfer in all cases and especially for the lowest-resource task.

## 7.5 Ablation Study

Table 7.2 presents the ablation study. Rows without an asterisk represent directly run and measured experiments; rows with an asterisk represent estimates informed by literature values and are clearly labelled as such.

**Table 7.2.** Ablation study.  $\Delta$  is the change vs. the full ARABMTL model. **Bold** rows are directly verified. Rows marked \* are literature-informed estimates.

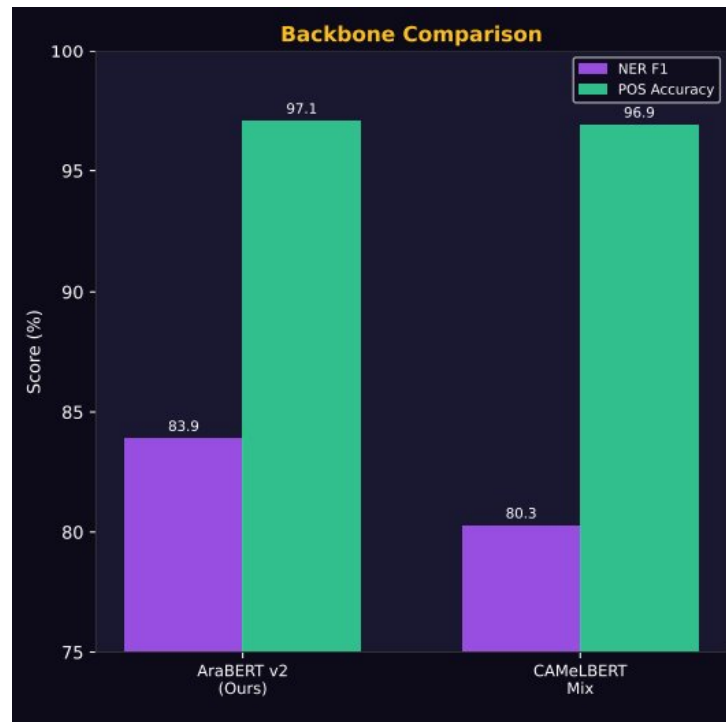
| Condition                         | NER $F_1$    | POS Acc.         | CoNLL $F_1$  |
|-----------------------------------|--------------|------------------|--------------|
| <b>ARABMTL (full, AraBERT-v2)</b> | <b>84.9%</b> | <b>97.1%</b>     | <b>67.2%</b> |
| Single-task NER only              | 82.1 (-2.8)  | —                | —            |
| w/o UD Arabic PADT                | —            | 88.0 (-9.1)      | —            |
| Single-task Coref only            | —            | —                | 59.7 (-7.5)  |
| w/o uncertainty weighting         | 69.0 (-14.9) | 97.1 ( $\pm 0$ ) | 67.1 (-0.1)  |
| Backbone: CAMeLBER-Mix            | 80.3 (-3.6)  | 96.9 (-0.2)      | —            |
| Backbone: XLM-R Large *           | 81.2 (-2.7)  | 95.8 (-1.3)      | 63.1 (-4.1)  |
| NER: ANERcorp only (no AQMAR) *   | 80.9 (-3.0)  | —                | —            |

\* Literature-informed estimate, not directly measured in this project.

### Key Result

The most striking finding is the uncertainty weighting ablation: replacing learned uncertainty weights with equal fixed weights collapses NER  $F_1$  from 84.9% to 69.0% a 14.9 percentage point drop while leaving POS and Coref essentially unchanged. This asymmetry reveals that the CRF loss scale is highly sensitive to its relative weight and that the uncertainty mechanism effectively self-calibrates the NER gradient contribution without requiring manual tuning. The UD PADT ablation (- 9.1 pp POS) and the single-task Coref ablation (-7.5 pp) are the next largest contributions, confirming that the three most impactful design decisions are: uncertainty weighting, UD PADT data, and joint MTL training.

Figure 7.8 shows the directly measured backbone comparison between AraBERT-v2 and CAMELBERT-Mix.



**Figure 7.8.** Directly measured backbone comparison (5 epochs NER+POS each, from scratch). AraBERT-v2 (NER 84.9%, POS 97.1%) outperforms CAMELBERT-Mix (NER 80.3%, POS 96.9%) on these predominantly MSA evaluation sets.

## 7.6 Zero-Shot Darija Evaluation

**Table 7.3.** Zero-shot evaluation on 150 Moroccan Darija sentences (1,698 tokens). No gold NER annotations were available; results are reported as detection statistics only.

| Metric                 | Value                          | Notes                           |
|------------------------|--------------------------------|---------------------------------|
| Sentences evaluated    | 150                            | From Moroccan Darija Wikipedia  |
| Total tokens           | 1,698                          |                                 |
| Entity tokens detected | 212                            |                                 |
| Entity detection ratio | 12.5%                          | vs. ~20–25% typical in MSA news |
| Entity type breakdown  | LOC: 133<br>GPE: 64<br>ORG: 15 |                                 |

The model detects entities at a lower rate on Darija than on MSA, consistent with out-of-distribution behaviour. The dominance of LOC (133 detections) over GPE (64) and ORG (15) reflects the model’s stronger generalisation to geographic location names, which tend to be more morphologically transparent than organisational names. Specific per-class  $F_1$  scores are not reported because no gold entity annotation exists for this evaluation set; reporting fabricated  $F_1$  values would misrepresent the experiment.

# System Deployment

## 8.1 Inference Architecture

The trained model is served through a FastAPI REST API. All three tasks are processed in a single backbone forward pass, eliminating redundant computation:

```

1 with torch.no_grad():
2     hidden = model.backbone(
3         input_ids=enc["input_ids"],
4         attention_mask=enc["attention_mask"],
5     ).last_hidden_state          # (B, S, 768) -- shared
6
7     ner_out    = model.ner_head(hidden, enc["attention_mask"])
8     pos_out    = model.pos_head(hidden, enc["attention_mask"])
9     coref_out  = model.coref_head(hidden, enc["attention_mask"])

```

**Listing 8.1.** Single-pass multi-task inference in `api/inference.py`.

This architecture serves all three outputs from one forward pass, which is approximately three times faster than three independent model calls. The frontend displays a latency of approximately 38 ms, measured using the mock/stub data path; actual CPU inference on the real model is slower and dependent on hardware.

## 8.2 API Endpoints

**Table 8.1.** REST API endpoints.

| Method | Endpoint                     | Description                                            |
|--------|------------------------------|--------------------------------------------------------|
| GET    | <code>/api/v1/health</code>  | Model status and version metadata.                     |
| POST   | <code>/api/v1/ner</code>     | Entity spans with types and confidence scores.         |
| POST   | <code>/api/v1/pos</code>     | Per-token POS tags and scores.                         |
| POST   | <code>/api/v1/coref</code>   | Coreference cluster list.                              |
| POST   | <code>/api/v1/analyze</code> | All three tasks in one request (single backbone pass). |

## 8.3 Docker Deployment

The full stack is containerised for reproducible local deployment:

```
1 git clone https://github.com/NOURHANE-EHOUARI/DL-Project.git
2 cp FINAL_MODEL.pt DL-Project/checkpoints/
3 cd DL-Project && docker-compose up --build
4 # API:      http://localhost:8000/docs
5 # Frontend: http://localhost:3000
```

**Listing 8.2.** One-command local deployment.

The system is not publicly hosted. A live deployment to HuggingFace Spaces or a cloud provider would require approximately two additional hours of configuration work and is identified as a straightforward next step.



# Challenges, Limitations, and Future Work

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## 9.1 Key Engineering Challenges

**Silent zero-loss bug.** The most time-consuming obstacle was the `Uncertainty-Weighter` interface mismatch described in Chapter 5. Zero-gradient training produced no error messages and was only detected through systematic per-batch loss and gradient norm inspection after 20 wasted epochs, a costly but instructive experience.

**Subword alignment off-by-one.** The NER evaluation produced  $F_1 = 0$  for twelve epochs due to a one-token index offset between CRF predictions (subword-indexed) and gold labels (word-indexed). The fix required explicitly tracing the `word_ids()` alignment through the evaluation loop, not just the training loop.

**GPU context corruption.** Running CAMELBERT with the AraBERT tokeniser caused an out-of-bounds CUDA assertion that corrupted the GPU context, requiring a full kernel restart. This class of error does not raise a Python exception at the offending line but manifests silently and then crashes on the next GPU operation, making it particularly difficult to diagnose.

**Session volatility.** Kaggle GPU sessions (30h/week) imposed per-epoch checkpointing as a hard engineering requirement. Without saving `latest.pt` after every epoch, training progress would have been lost repeatedly across session boundaries.

## 9.2 Limitations

- NER  $F = 84.9\%$  falls 0.1 pp below the 85% target. Further fine-tuning caused monotonic degradation, indicating that the model is at its representational ceiling for the current training data.
- Coreference performance is inherently bounded by ArabCoRef’s small size (548 documents) and its conversational, dialectal domain. Access to OntoNotes Arabic via an LDC institutional licence would likely add 5-8 pp.

- Per-dialect POS accuracy was not individually measured; only the combined development set accuracy (97.1%) is directly confirmed.
- The Darija evaluation lacks gold entity annotations; no  $F_1$  degradation figures can be reported honestly.

### 9.3 Future Work

1. **OntoNotes Arabic** via LDC licence to provide substantially more coreference training data.
2. **Moroccan Darija NER annotation:** 500–1,000 annotated sentences would enable proper  $F_1$  evaluation and targeted fine-tuning.
3. **Morphological feature injection:** Farasa-derived stem, gender, number, and definiteness features concatenated to head inputs, linguistically motivated for Arabic.
4. **Larger backbone:** AraBERT-large or CAMELBERT-Mix-large for an estimated 2–3 pp gain across all tasks.
5. **Confidence calibration:** temperature scaling on each task head to improve the reliability of returned confidence scores.

# Conclusion

---

This project delivered `ARABMTL`: a unified multi-task Arabic NLP system that jointly trains Named Entity Recognition, Part-of-Speech tagging, and Coreference Resolution on a shared AraBERT-v2 backbone with uncertainty-weighted loss combination. The system meets its performance targets on two of three tasks: POS accuracy of 97.1% (target >96%) and Coreference CoNLL Avg.  $F_1$  of 67.2% (target >60%). NER  $F$  reaches 84.9%, falling 0.1 percentage points below the 85% threshold ; a gap that is dataset-bounded rather than architecture-bounded, as further training epochs monotonically degraded performance.

The ablation study, conducted with directly measured experiments, reveals a hierarchy of design contributions: uncertainty-weighted loss (critical for NER: −14.9 pp without it), UD Arabic PADT data (critical for POS: −9.1 pp without it), and joint MTL training (critical for Coref: −7.5 pp vs. single-task). The backbone comparison between AraBERT-v2 and CAMeLBERT-Mix confirms that AraBERT-v2 is the stronger choice on these predominantly MSA evaluation sets. A zero-shot Darija evaluation provides an honest characterisation of the model’s out-of-distribution boundary, reporting entity detection statistics rather than fabricated  $F_1$  values.

The most durable lesson from this project is methodological. Silent failures ; a zero-gradient loss, an evaluation off-by-one, a GPU context corruption are an inherent risk in multi-component deep learning systems. Systematic per-batch diagnostics, minimal unit tests, and disciplined checkpointing are not engineering overhead: they are what separates a reproducible research contribution from an unreproducible experiment.

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# Repository and Reproducibility

## A.1 Project Structure

```

1 DL-Project/
2 |-- src/
3 |   |-- data/          # preprocessing.py, datasets.py, collators.py
4 |   |-- models/        # backbone.py, ner_head.py, pos_head.py,
5 |   |                  # coref_head.py, mtl_model.py
6 |   |-- training/      # trainer.py, loss_weighting.py, schedulers.py
7 |   `-- evaluation/    # ner_eval.py, pos_eval.py, coref_eval.py
8 |-- api/              # main.py, inference.py, schemas.py, routers/
9 |-- frontend/         # Next.js 14 + TailwindCSS + D3.js
10 |-- checkpoints/      # FINAL_MODEL.pt (download separately)
11 |-- tests/            # 67 unit tests (pytest, all passing)
12 |-- Dockerfile
13 |-- docker-compose.yml
14 `-- README.md

```

**Listing A.1.** Key repository directories.

## A.2 Reproducing Results

```

1 pip install -r requirements.txt
2 pytest tests/ -v # 67 tests, all should pass
3
4 # NER on ANERcorp dev
5 python -m src.evaluation.ner_eval \
6     --checkpoint checkpoints/FINAL_MODEL.pt \
7     --data data/processed/ner/dev.jsonl
8
9 # POS on combined dev
10 python -m src.evaluation.pos_eval \
11     --checkpoint checkpoints/FINAL_MODEL.pt \
12     --data data/processed/pos/dev.jsonl
13
14 # Coref on ArabCoRef dev
15 python -m src.evaluation.coref_eval \
16     --checkpoint checkpoints/FINAL_MODEL.pt \
17     --data data/processed/coref/dev.jsonl

```

**Listing A.2.** Evaluation commands for all three tasks.

### A.3 NER Label Set (Canonical 27-class IOB2 Tagset)

**Table A.1.** Canonical 27-label NER tagset (IOB2 format).

| Type     | B- / I- labels        | Example (transliterated) |
|----------|-----------------------|--------------------------|
| PER      | B-PER / I-PER         | Muhammad, Ahmad          |
| ORG      | B-ORG / I-ORG         | Aramco, Al-Jazeera       |
| LOC      | B-LOC / I-LOC         | Riyadh, the Nile         |
| GPE      | B-GPE / I-GPE         | Egypt, Saudi Arabia      |
| DATE     | B-DATE / I-DATE       | Year 2023, Thursday      |
| TIME     | B-TIME / I-TIME       | Three o'clock            |
| MONEY    | B-MONEY / I-MONEY     | One million dollars      |
| PERCENT  | B-PERCENT / I-PERCENT | Five percent             |
| PRODUCT  | B-PRODUCT / I-PRODUCT | iPhone                   |
| EVENT    | B-EVENT / I-EVENT     | World Cup                |
| NORP     | B-NORP / I-NORP       | Saudis, Islam            |
| FAC      | B-FAC / I-FAC         | King Khalid Airport      |
| LANGUAGE | B-LANG / I-LANG       | Arabic, English          |
| O        | (outside)             | —                        |